Newman-Penrose Formalism Calculations

Stephani et al., Equation (14.41)

Metric

$$ds^2 = -\mathrm{d}t \otimes \mathrm{d}t + t^{\frac{6}{5}} e^{\left(\frac{8}{15}\,\sqrt{3}\sqrt{2}\sqrt{b^2+1}x\right)} \mathrm{d}z \otimes \mathrm{d}z + \left(b^2+1\right) t^2 \mathrm{d}x \otimes \mathrm{d}x + b t^{\frac{6}{5}} e^{\left(\frac{2}{15}\,\sqrt{3}\sqrt{2}\sqrt{b^2+1}x\right)} \mathrm{d}x \otimes \mathrm{d}y + b t^{\frac{6}{5}} e^{\left(\frac{2}{15}\,\sqrt{3}\sqrt{2}\sqrt{b^2+1}x\right)} \mathrm{d}y \otimes \mathrm{d}x + t^{\frac{2}{5}} e^{\left(\frac{4}{15}\,\sqrt{3}\sqrt{2}\sqrt{b^2+1}x\right)} \mathrm{d}y \otimes \mathrm{d}y$$

Coordinates

$$(x^\mu) = (t, z, x, y)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\,\sqrt{2}\right)\,dt + \left(\frac{1}{2}\,\sqrt{2} t^{\frac{3}{5}} e^{\left(\frac{4}{15}\,\sqrt{6}\sqrt{b^2+1}x\right)}\right)\,dz$$

$$n = \left(\frac{1}{2}\,\sqrt{2}\right)\,dt + \left(-\frac{1}{2}\,\sqrt{2} t^{\frac{3}{5}} e^{\left(\frac{4}{15}\,\sqrt{6}\sqrt{b^2+1}x\right)}\right)\,dz$$

$$m = \left(\frac{1}{2}\,\sqrt{2}(i\,b+1)t\right)\,dx + \left(\frac{1}{2} i\,\sqrt{2} t^{\frac{1}{5}} e^{\left(\frac{2}{15}\,\sqrt{6}\sqrt{b^2+1}x\right)}\right)\,dy$$

$$\overline{m} = \left(\frac{1}{2}\,\sqrt{2}(-i\,b+1)t\right)\,dx + \left(-\frac{1}{2} i\,\sqrt{2} t^{\frac{1}{5}} e^{\left(\frac{2}{15}\,\sqrt{6}\sqrt{b^2+1}x\right)}\right)\,dy$$

Contravariant components

$$l = \left(-\frac{1}{2}\,\sqrt{2}\right)\,dt + \left(\frac{\sqrt{2} e^{\left(-\frac{4}{15}\,\sqrt{3}\sqrt{2}\sqrt{b^2+1}x\right)}}{2\,t^{\frac{3}{5}}}\right)\,dz$$

$$n = \left(-\frac{1}{2}\,\sqrt{2}\right)\,dt + \left(-\frac{\sqrt{2}e^{\left(-\frac{4}{15}\,\sqrt{3}\sqrt{2}\sqrt{b^2+1}x\right)}}{2\,t^{\frac{3}{5}}}\right)\,dz$$

$$m = \left(\frac{\sqrt{2}}{2\,t}\right)\,dx + \left(-\frac{\sqrt{2}(b-i)e^{\left(-\frac{2}{15}\,\sqrt{3}\sqrt{2}\sqrt{b^2+1}x\right)}}{2\,t^{\frac{1}{5}}}\right)\,dy$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2\,t}\right)\,dx + \left(-\frac{\sqrt{2}(b+i)e^{\left(-\frac{2}{15}\,\sqrt{3}\sqrt{2}\sqrt{b^2+1}x\right)}}{2\,t^{\frac{1}{5}}}\right)\,dy$$

Directional Derivatives

$$D = -\frac{1}{2}\,\sqrt{2}\,\partial_t + \frac{\sqrt{2}e^{\left(-\frac{4}{15}\,\sqrt{3}\sqrt{2}\sqrt{b^2+1}x\right)}}{2\,t^{\frac{3}{5}}}\,\partial_z$$

$$\Delta = -\frac{1}{2}\,\sqrt{2}\,\partial_t - \frac{\sqrt{2}e^{\left(-\frac{4}{15}\,\sqrt{3}\sqrt{2}\sqrt{b^2+1}x\right)}}{2\,t^{\frac{3}{5}}}\,\partial_z$$

$$\delta = \frac{\sqrt{2}}{2\,t}\,\partial_x - \frac{\sqrt{2}(b-i)e^{\left(-\frac{2}{15}\,\sqrt{3}\sqrt{2}\sqrt{b^2+1}x\right)}}{2\,t^{\frac{1}{5}}}\,\partial_y$$

$$\overline{\delta} = \frac{\sqrt{2}}{2\,t}\,\partial_x - \frac{\sqrt{2}(b+i)e^{\left(-\frac{2}{15}\,\sqrt{3}\sqrt{2}\sqrt{b^2+1}x\right)}}{2\,t^{\frac{1}{5}}}\,\partial_y$$

Spin Coefficients

$$\rho = \frac{3\,\sqrt{2}}{10\,t}$$

$$\sigma = \frac{\sqrt{2}(i\,b+1)}{5\,t}$$

$$\kappa = \frac{2\,\sqrt{3}\sqrt{b^2+1}}{15\,t}$$

$$\mu = -\frac{3\,\sqrt{2}}{10\,t}$$

$$\lambda = \frac{\sqrt{2}(i\,b-1)}{5\,t}$$

$$\nu = -\frac{2\,\sqrt{3}\sqrt{b^2+1}}{15\,t}$$

$$\alpha = -\frac{\sqrt{3}\sqrt{b^2+1}}{15\,t}$$

$$\beta = \frac{\sqrt{3}\sqrt{b^2+1}}{15\,t}$$

$$\tau = -\frac{2\,\sqrt{3}\sqrt{b^2+1}}{15\,t}$$

$$\pi = \frac{2\,\sqrt{3}\sqrt{b^2+1}}{15\,t}$$

$$\epsilon = \frac{\sqrt{2}(-2i\,b-3)}{20\,t}$$

$$\gamma = \frac{\sqrt{2}(-2i\,b+3)}{20\,t}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{3\,(2\,b+1)(2\,b-1)}{50\,t^2}$$

$$\Phi_{01} = -\frac{4\,\sqrt{3}\sqrt{2}\sqrt{b^2+1}(i\,b+1)}{75\,t^2}$$

$$\Phi_{02} = \frac{8\left(b^2+1\right)}{75\,t^2}$$

$$\Phi_{10} = \frac{4\,\sqrt{3}\sqrt{2}\sqrt{b^2+1}(i\,b-1)}{75\,t^2}$$

$$\Phi_{11} = -\frac{4\,b^2-5}{60\,t^2}$$

$$\Phi_{12} = -\frac{4\,\sqrt{3}\sqrt{2}\sqrt{b^2+1}(i\,b+1)}{75\,t^2}$$

$$\Phi_{20} = \frac{8\left(b^2+1\right)}{75\,t^2}$$

$$\Phi_{21} = \frac{4\,\sqrt{3}\sqrt{2}\sqrt{b^2+1}(i\,b-1)}{75\,t^2}$$

$$\Phi_{22} = -\frac{3\,(2\,b+1)(2\,b-1)}{50\,t^2}$$

$$\Lambda = -\frac{44\,b^2+17}{900\,t^2}$$

Weyl Tensor Components

$$\Psi_0 = -\frac{2\left(8\,b^2 - 9i\,b - 1\right)}{75\,t^2}$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{4\left(b^2 + 1\right)}{225\,t^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = -\frac{2\left(8\,b^2 + 9i\,b - 1\right)}{75\,t^2}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"